THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** normal	Girls often grow earlier.
2	**(c)** growth spurt	Both grow taller.
3	**(d)** neck	Front of the neck.
4	**(b)** larynx is growing	Voice box.
5	**(d)** fast growth needs more energy	Energy.
6	**(b)** abdomen	Near the stomach.
7	**(a)** goitre	Thyroid.
8	**(d)** fruit and nuts	Healthy.
9	**(d)** kidneys	Small glands.
10	**(a)** diabetes	Sugar high.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Larynx grows more in boys (1); Testosterone causes this (1).	1+1
12	Strong muscles and bones (1); Better mood (1).	1+1
13	Part of the body's cycle (1); Prepares for possible pregnancy (1).	1+1
14	It releases hormones that stimulate testes or ovaries (1); It controls growth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Obesity risk (1); Low vitamins and minerals (1); Add fruits, vegetables, milk (1).	1+1+1
16	Peer pressure (1); Curiosity (1); Awareness and refusal skills (1).	1+1+1
17	Oil glands active (1); Dead cells block pores (1); Bacteria (1).	1+1+1
	OR: Egg matures (1); Lining thickens (1); If no fertilisation, lining sheds (1).	1+1+1
18	Endocrine: no ducts (1); Digestive: ducts carry juices (1); Examples: thyroid vs salivary (1).	1+1+1
	OR: Guidance (1); Reduces stress (1); Prevents wrong steps (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Boys: testosterone (1); Voice, beard (1); Girls: oestrogen (1); Breast, periods (1); Both: growth spurt (1).	1+1+1+1+1
	OR: Mood swings (1); Need for acceptance (1); Talk to family (1); Hobbies (1); Seek help if needed (1).	1+1+1+1+1
20	Nutrition for growth (1); Sleep for rest (1); Hygiene for health (1); Exercise for fitness (1); All support mental health (1).	1+1+1+1+1
	OR: Iodine: goitre, iodised salt (1); Iron: anaemia, green vegetables (1); Insulin: diabetes, diet control (1); Regular checks (1); Medical care (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that puberty starts at the same age in all children; it starts **earlier in girls** than in boys and varies from person to person.
- Writing that hormones are carried by ducts; endocrine glands are **ductless** and hormones travel in the blood.
- Confusing the glands: *testes* make testosterone, *ovaries* make oestrogen, *pancreas* makes insulin.
- Saying the voice breaks because of the throat or lungs; it is due to the enlargement of the **voice box (larynx)**.
- Calling menstruation an illness; it is a natural process and needs hygiene, not shame.
- Treating mood swings as bad behaviour; they are caused by hormones and can be managed with sleep, exercise and talking.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** loss of milk teeth at 5	Teeth shedding starts earlier.
2	**(d)** brain	Brain.
3	**(a)** iron deficiency	Anaemia.
4	**(d)** talk to a trusted adult or counsellor	Seek help.
5	**(d)** haemoglobin	Haemoglobin is a protein in blood.
6	**(b)** testosterone	Male hormone.
7	**(d)** regular exercise	Healthy way.
8	**(d)** salivary gland	It has ducts.
9	**(b)** oestrogen	Female hormone.
10	**(a)** to prevent infection and stay clean	Hygiene.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hormones are chemical messengers made by glands (1); Enzymes help in reactions (1).	1+1
12	Plan and rest (1); Talk to elders (1).	1+1
13	Iodine is needed for thyroxine (1); Prevents goitre (1).	1+1
14	Each person grows at own rate (1); Comparison can cause stress (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Height, voice, hair (1); Mood swings (1); Curiosity and independence (1).	1+1+1
16	Diabetes (1); Goitre (1); Growth problems (1).	1+1+1
17	Growth: increase in size (1); Development: new abilities (1); Both occur in adolescence (1).	1+1+1
	OR: Different hormones (1); Different rates (1); Normal variation (1).	1+1+1
18	Strangers may mislead (1); Privacy risk (1); Tell elders (1).	1+1+1
	OR: Heart beats faster (1); Breathing quickens (1); More blood to muscles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pituitary: growth (1); Thyroid: metabolism (1); Pancreas: sugar (1); Adrenal: emergency (1); Governed by hormones (1).	1+1+1+1+1
	OR: Breakfast, lunch, dinner balanced (1); Study time (1); One hour exercise (1); 8 hours sleep (1); Free time for hobbies (1).	1+1+1+1+1
20	Health problems (1); Financial burden (1); Crime risk (1); Awareness (1); Counselling and treatment (1).	1+1+1+1+1
	OR: Benefits: learning (1); Friends (1); Cautions: privacy (1); Time limits (1); Respect (1).	1+1+1+1+1

Mistakes to avoid:

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- Confusing the glands: *testes* make testosterone, *ovaries* make oestrogen, *pancreas* makes insulin.
- Saying the voice breaks because of the throat or lungs; it is due to the enlargement of the **voice box (larynx)**.
- Calling menstruation an illness; it is a natural process and needs hygiene, not shame.
- Treating mood swings as bad behaviour; they are caused by hormones and can be managed with sleep, exercise and talking.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Adolescence: A Stage of Growth and Change** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** menarche	Starts in girls.
2	**(c)** pituitary	Master gland.
3	**(b)** saying no firmly	Make a healthy decision.
4	**(a)** growth of beard	Beard is in boys.
5	**(b)** adrenaline	Emergency hormone.
6	**(a)** roti and rice	Carbohydrates.
7	**(d)** ovary	Female gonad.
8	**(b)** testes and ovaries	Reproductive organs.
9	**(a)** milk	Bone growth.
10	**(c)** testes	Male gonad.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	High fat and salt (1); Low nutrients (1).	1+1
12	Their pancreas makes too little (1); It controls blood sugar (1).	1+1
13	Growth hormone is released (1); Helps the brain and body to rest (1).	1+1
14	Increases heartbeat (1); Prepares for action (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Thyroxine controls energy use (1); Needs iodine (1); Deficiency causes goitre (1).	1+1+1
16	Counsellors (1); Sports (1); Fair workload (1).	1+1+1
17	Damages lungs (1); Addictive (1); Reduces stamina (1).	1+1+1
	OR: Talk to friends and elders (1); Hobbies (1); Sleep and exercise (1).	1+1+1
18	Iron loss (1); Energy needs (1); Prevents weakness (1).	1+1+1
	OR: Support (1); Guidance (1); Positive habits (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cycle length 28 days (1); Bleeding for 3 to 5 days (1); Hygiene products (1); Iron rich food (1); Rest (1).	1+1+1+1+1
	OR: Equal opportunity (1); No teasing (1); Safe space (1); Confidence (1); Teamwork (1).	1+1+1+1+1
20	Myths cause fear (1); Wrong practices (1); Teachers and doctors (1); Books (1); Talk openly (1).	1+1+1+1+1
	OR: Fast growth (1); More protein and calcium (1); Milk, dal, eggs (1); Green vegetables (1); Whole grains (1).	1+1+1+1+1

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